

The 26th Privacy Enhancing Technologies Symposium



July 20–25, 2026, Calgary, Canada



Frequency Estimation of **Correlated** **Multi-attribute Data** under Local Differential Privacy

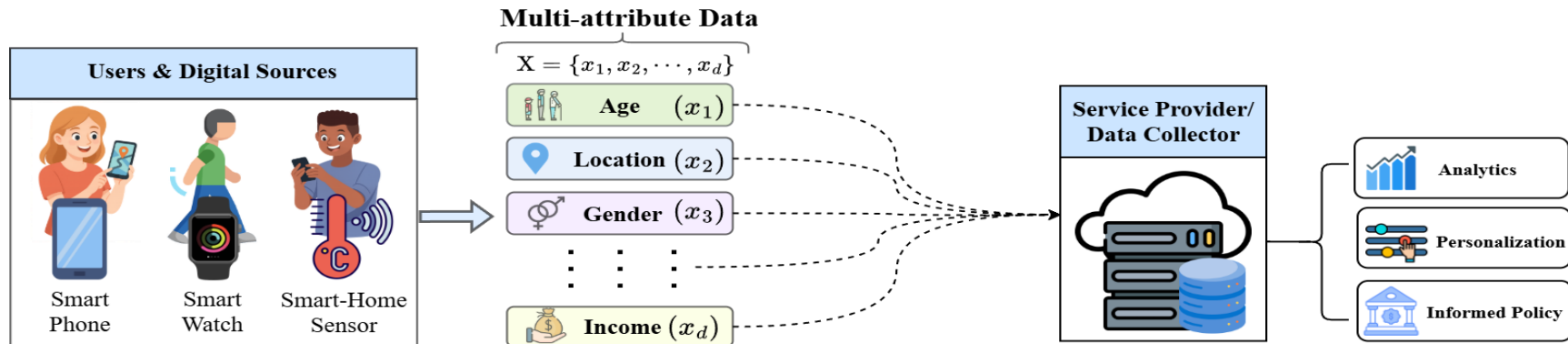
Authors: Shafizur Rahman Seeam, Ye Zheng, Yidan Hu

RIT

**Rochester Institute
of Technology**

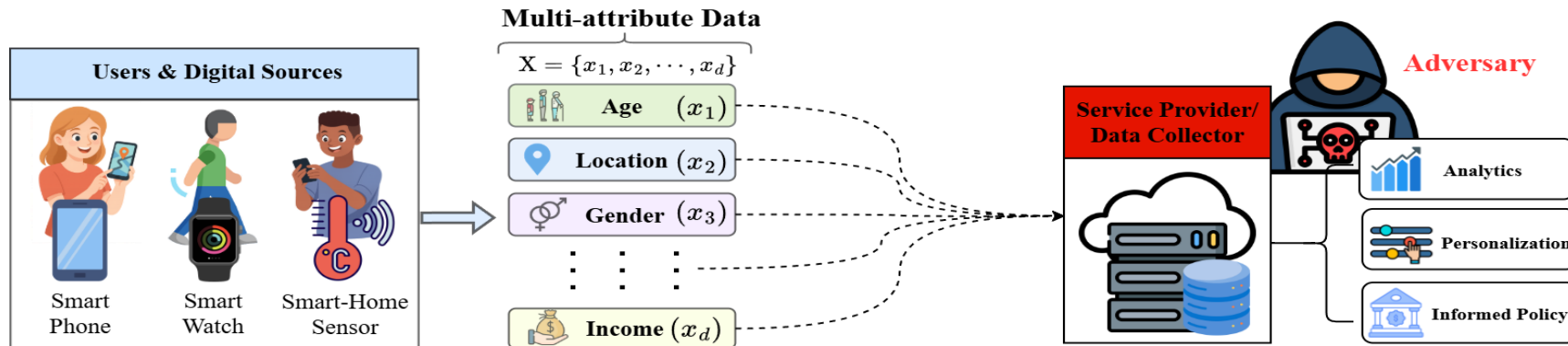
Multi-Attribute Data Collection

Modern connected devices (e.g., IoT, smart watch) **collect multiple attributes per user**, enabling better policymaking and analytics for providers, and personalization for users.



Multi-Attribute Data Collection

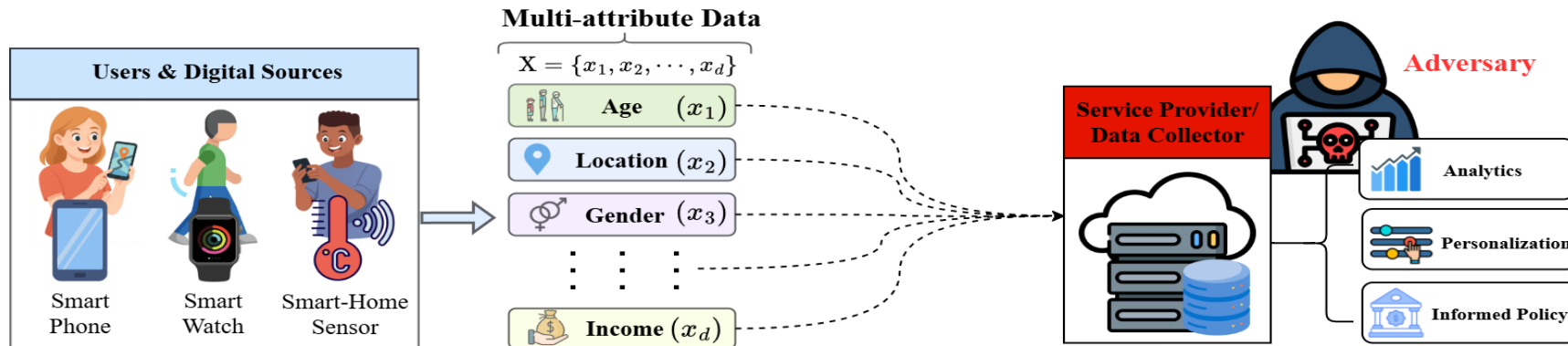
Modern connected devices (e.g., IoT, smart watch) **collect multiple attributes per user**, enabling better policymaking and analytics for providers, and personalization for users.



- If the data collector is untrusted or compromised, multi-attribute records may expose sensitive information (e.g., age and income), thereby increasing the risk of individual re-identification.

Multi-Attribute Data Collection

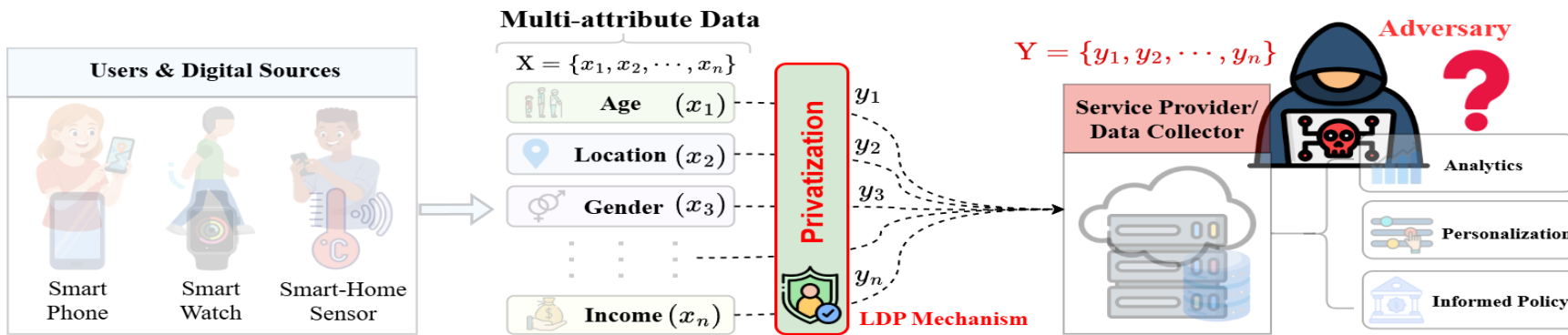
Modern connected devices (e.g., IoT, smart watch) **collect multiple attributes per user**, enabling better policymaking and analytics for providers, and personalization for users.



- If the data collector is **untrusted or compromised**, multi-attribute records may expose sensitive information (e.g., age and income), thereby increasing the risk of individual re-identification.
- Therefore, user data should be **privatized before leaving the device**, protecting individual users while still enabling the server to compute privacy-preserving statistics.

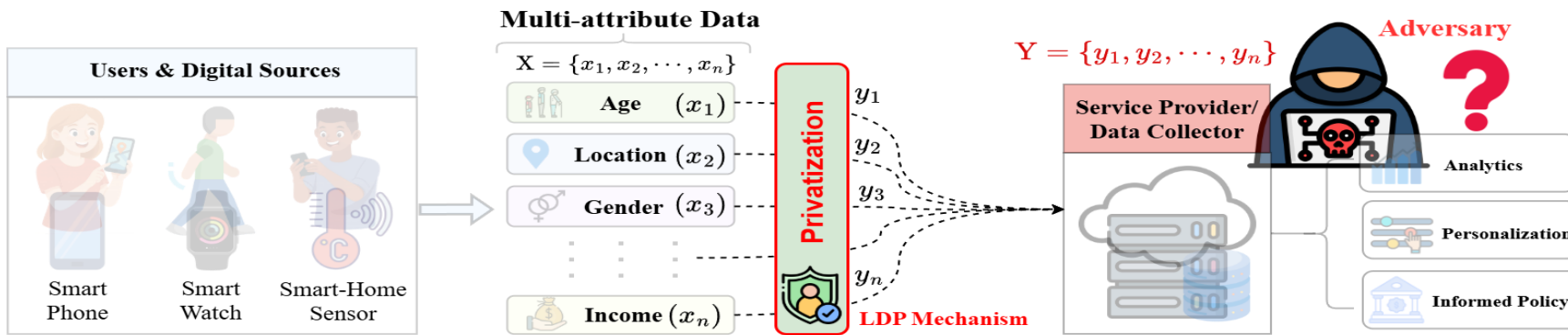
Multi-Attribute Data Collection under Local Differential Privacy (LDP)

LDP is a widely used framework for privacy-preserving data collection, adopted by Google, Apple, and Microsoft. Under LDP, each user randomizes their record on-device as $Y := M(X)$, where M denotes an LDP mechanism, and the server observes only the privatized reports.



Multi-Attribute Data Collection under Local Differential Privacy (LDP)

LDP is a widely used framework for privacy-preserving data collection, adopted by Google, Apple, and Microsoft. Under LDP, each user randomizes their record on-device as $Y := M(X)$, where M denotes an LDP mechanism, and the server observes only the privatized reports.

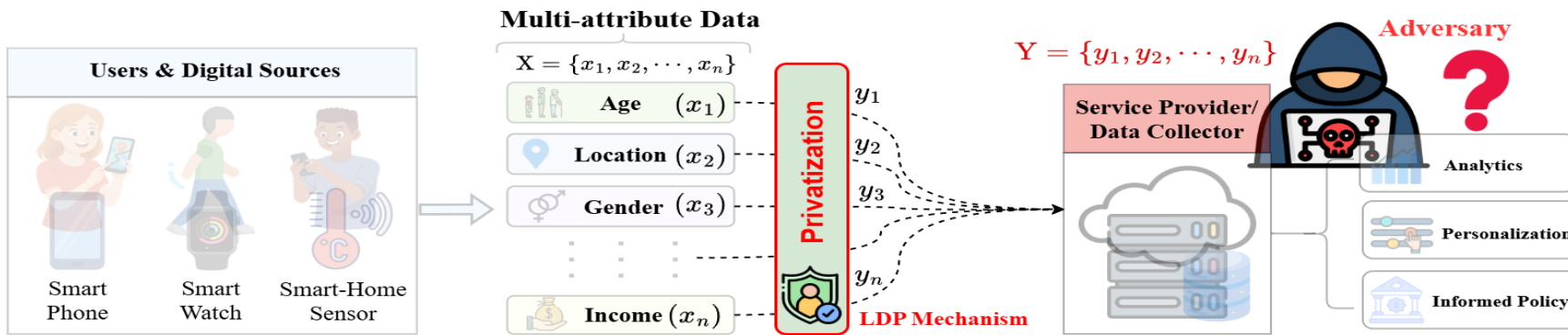


- **Formal Guarantee:** For any two possible inputs x, x' , and any output y

$$\Pr[M(x) = y] \leq e^{\epsilon} \Pr[M(x') = y]$$

Multi-Attribute Data Collection under Local Differential Privacy (LDP)

LDP is a widely used framework for privacy-preserving data collection, adopted by Google, Apple, and Microsoft. Under LDP, each user randomizes their record on-device as $Y := M(X)$, where M denotes an LDP mechanism, and the server observes only the privatized reports.



- **Formal Guarantee:** For any two possible inputs x, x' , and any output y

$$\Pr[M(x) = y] \leq e^{\epsilon} \Pr[M(x') = y]$$

- Privacy budget : A smaller ϵ provides stronger privacy at the cost of lower utility.

Existing Multi-Attribute LDP Mechanisms and their Limitations

Frequency estimation (estimating how often values appears in a population) is well studied for single-attribute data under LDP. However, extending LDP to multi-attribute records remains challenging.

Existing Multi-Attribute LDP Mechanisms and their Limitations

Frequency estimation (estimating how often values appears in a population) is well studied for single-attribute data under LDP. However, extending LDP to multi-attribute records remains challenging.

- Split Budget (SPL): Each attribute is perturbed using a privacy budget of ϵ/d .

User	x_1	x_2	...	x_d
u_1	ϵ/d	ϵ/d	ϵ/d	ϵ/d
u_2	ϵ/d	ϵ/d	ϵ/d	ϵ/d

Existing Multi-Attribute LDP Mechanisms and their Limitations

Frequency estimation (estimating how often values appears in a population) is well studied for single-attribute data under LDP. However, extending LDP to multi-attribute records remains challenging.

- Split Budget (SPL): Each attribute is perturbed using a privacy budget of ϵ/d .

User	x_1	x_2	...	x_d
u_1	ϵ/d	ϵ/d	ϵ/d	ϵ/d
u_2	ϵ/d	ϵ/d	ϵ/d	ϵ/d

- Each attribute receives a small privacy budget, leading to high noise and poor frequency estimates.

Existing Multi-Attribute LDP Mechanisms and their Limitations

Frequency estimation (estimating how often values appears in a population) is well studied for single-attribute data under LDP. However, **extending LDP to multi-attribute records remains challenging.**

- Split Budget (SPL): Each attribute is perturbed using a privacy budget of ϵ/d .

User	x_1	x_2	...	x_d
u_1	ϵ/d	ϵ/d	ϵ/d	ϵ/d
u_2	ϵ/d	ϵ/d	ϵ/d	ϵ/d

- Each attribute receives a small privacy budget, leading to **high noise and poor frequency estimates.**

- Random Sampling + (Realistic) Fake Data (RS+FD/ RS+RFD): Perturb one randomly selected attribute with full ϵ , then generate (realistic) fake values for the remaining attributes.

User	x_1	x_2	...	x_d
u_1	fake	ϵ	fake	fake
u_2	ϵ	fake	fake	fake

Existing Multi-Attribute LDP Mechanisms and their Limitations

Frequency estimation (estimating how often values appears in a population) is well studied for single-attribute data under LDP. However, **extending LDP to multi-attribute records remains challenging.**

- Split Budget (SPL): Each attribute is perturbed using a privacy budget of ϵ/d .

User	x_1	x_2	...	x_d
u_1	ϵ/d	ϵ/d	ϵ/d	ϵ/d
u_2	ϵ/d	ϵ/d	ϵ/d	ϵ/d

- Each attribute receives a small privacy budget, leading to **high noise and poor frequency estimates.**

- Random Sampling + (Realistic) Fake Data (RS+FD/ RS+RFD): Perturb one randomly selected attribute with full ϵ , then generate (realistic) fake values for the remaining attributes.

User	x_1	x_2	...	x_d
u_1	fake	ϵ	fake	fake
u_2	ϵ	fake	fake	fake

- Most reported values are synthetic; **RS+FD** suffers from **inaccurate uniform fake data**, while **RS+RFD** **relies on external priors** to generate realistic fake data.

Existing Multi-Attribute LDP Mechanisms and their Limitations

Frequency estimation (estimating how often values appears in a population) is well studied for single-attribute data under LDP. However, **extending LDP to multi-attribute records remains challenging.**

- Split Budget (SPL): Each attribute is perturbed using a privacy budget of ϵ/d .

User	x_1	x_2	...	x_d
u_1	ϵ/d	ϵ/d	ϵ/d	ϵ/d
u_2	ϵ/d	ϵ/d	ϵ/d	ϵ/d

- Each attribute receives a small privacy budget, leading to **high noise and poor frequency estimates.**

- Random Sampling + (Realistic) Fake Data (RS+FD/ RS+RFD): Perturb one randomly selected attribute with full ϵ , then generate (realistic) fake values for the remaining attributes.

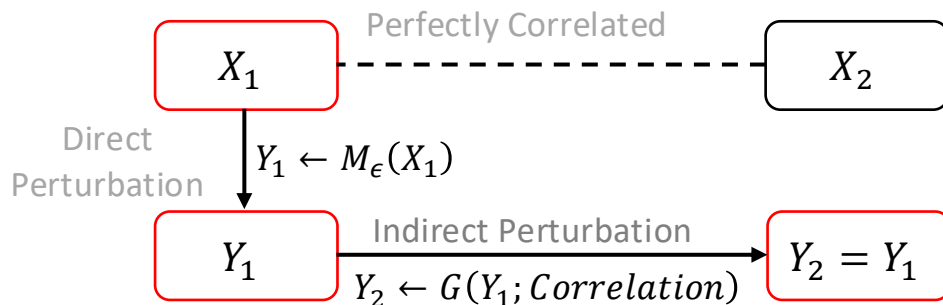
User	x_1	x_2	...	x_d
u_1	fake	ϵ	fake	fake
u_2	ϵ	fake	fake	fake

- Most reported values are synthetic; **RS+FD** suffers from **inaccurate uniform fake data**, while **RS+RFD** relies on external priors to generate realistic fake data.

- Key Observation: Most existing methods **treat attributes independently.**
 - However, **attributes in real-world data are often correlated** (e.g., Age $\leftrightarrow$ Salary)

Key Idea: **Exploit Correlations** Among Attributes to Improve Utility

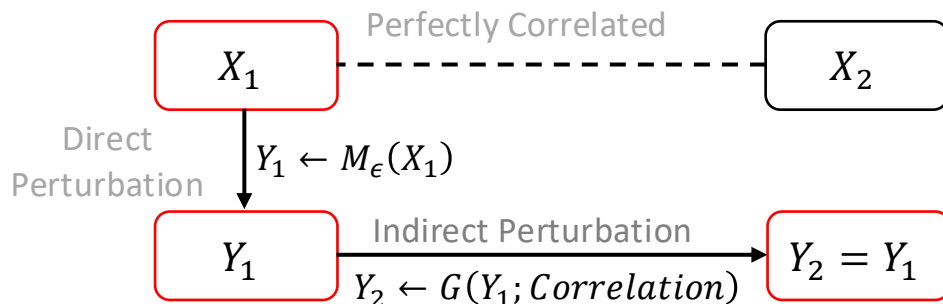
For two perfectly correlated attributes ($X_1 = X_2$), randomly select one attribute and perturb it using the full privacy budget ϵ . The other privatized attribute can then be generated as $Y_2 = Y_1$



- G operates only on the perturbed value
 - post-processing
 - **no extra privacy cost, still ϵ -LDP**

Key Idea: **Exploit Correlations** Among Attributes to Improve Utility

For two perfectly correlated attributes ($X_1 = X_2$), randomly select one attribute and perturb it using the full privacy budget ϵ . The other privatized attribute can then be generated as $Y_2 = Y_1$



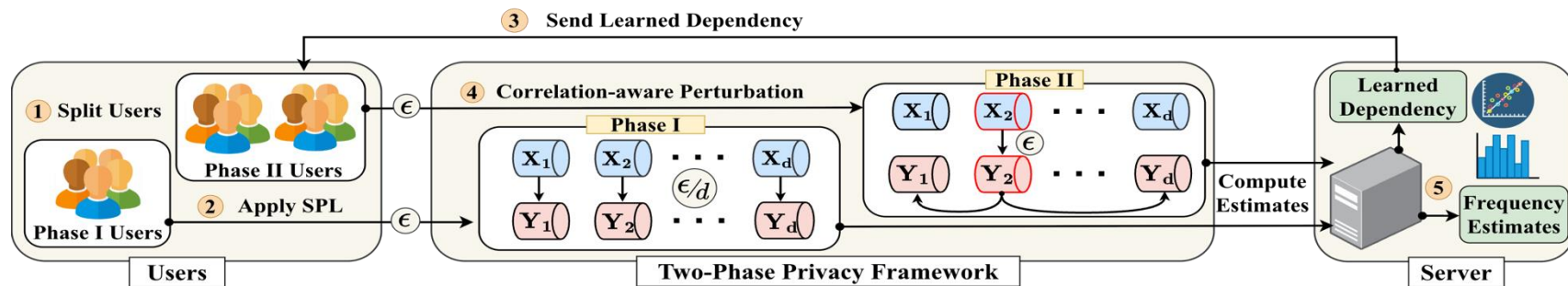
- G operates only on the perturbed value
 - post-processing
 - **no extra privacy cost, still ϵ -LDP**

Two challenges:

- **Correlations are imperfect**: $X_1 = X_2$ does not generally hold.
- **Correlations are unknown**: raw data cannot be directly used for correlation estimation.

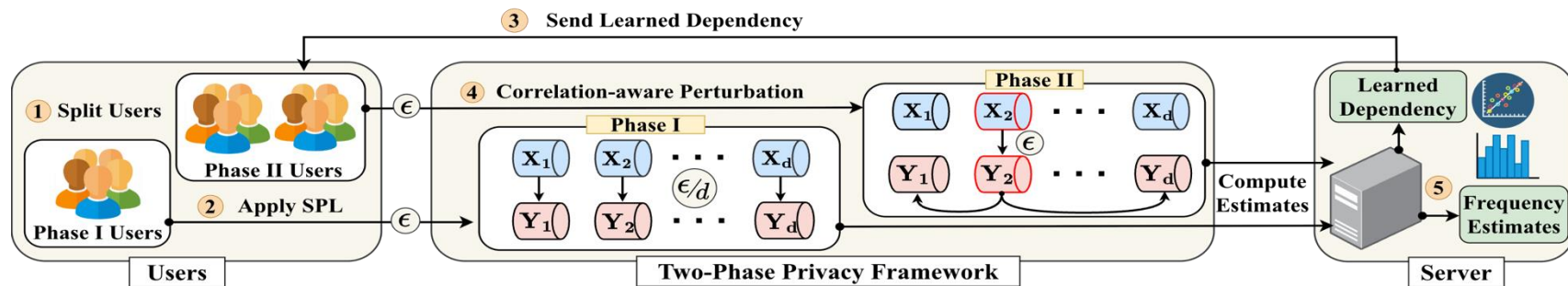
Correlated Randomized Response (Corr-RR)

We introduce a novel **two-phase privacy framework** and instantiate it with **Corr-RR**.



Correlated Randomized Response (Corr-RR)

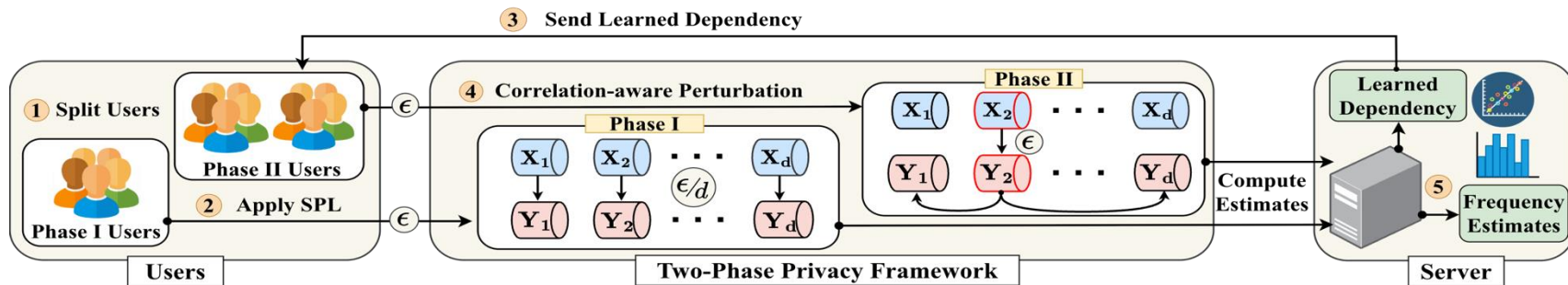
We introduce a novel **two-phase privacy framework** and instantiate it with **Corr-RR**.



- Phase I: A small subset of users perturbs all attributes using SPL (ϵ/d). From these privatized reports, the server estimates noisy marginal distributions.
 - Convert **estimates into reuse probabilities p_y** for Phase II.

Correlated Randomized Response (Corr-RR)

We introduce a novel **two-phase privacy framework** and instantiate it with **Corr-RR**.



- Phase I: A small subset of users perturbs all attributes using SPL (ϵ/d). From these privatized reports, the server estimates noisy marginal distributions.
 - Convert **estimates into reuse probabilities p_y** for Phase II.
- Phase II: Each remaining user perturbs one random pivot attribute with full ϵ . **Each non-pivot attribute reuses the pivot output with probability p_y** , otherwise, it samples another value
 - Non-pivot reports are generated by **post-processing the pivot (no extra privacy cost)**.

Choosing the Reuse Probability p_y

p_y is the probability that a non-pivot attribute reuses the pivot's privatized output.

$$\Pr(Y_j = Y_s) = p_y, \Pr(Y_j \neq Y_s) = 1 - p_y$$

Choosing the Reuse Probability p_y

p_y is the probability that a non-pivot attribute reuses the pivot's privatized output.

$$\Pr(Y_j = Y_s) = p_y, \Pr(Y_j \neq Y_s) = 1 - p_y$$

- Corr-RR **selects p_y by minimizing the estimator's overall MSE**, which has a closed-form solution.

(Proposition 1 in the paper)

$$p_y = \arg \min_{p_y \in [0,1]} MSE_{avg}(p_y)$$

Choosing the Reuse Probability p_y

p_y is the probability that a non-pivot attribute reuses the pivot's privatized output.

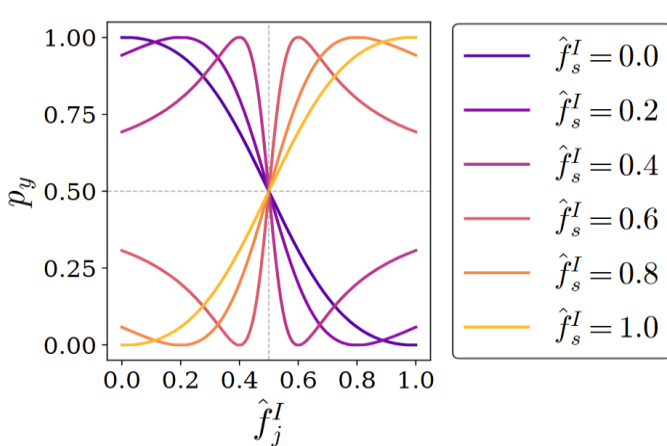
$$\Pr(Y_j = Y_s) = p_y, \Pr(Y_j \neq Y_s) = 1 - p_y$$

- Corr-RR **selects p_y by minimizing the estimator's overall MSE**, which has a closed-form solution.

(Proposition 1 in the paper)

$$p_y = \arg \min_{p_y \in [0,1]} MSE_{avg}(p_y)$$

- True marginals are unknown**; Corr-RR plugs in Phase I estimates:



$$f_j(v), f_s(v) \approx \hat{f}_j^I(v), \hat{f}_s^I(v)$$

☒ **Similar Phase I marginals $\rightarrow$ Higher Reuse.**

☐ **Different Phase I marginals $\rightarrow$ Lower Reuse.**

Figure: How the p_y changes with the Phase I marginal estimates.

Privacy Guarantee of Corr-RR

Corr-RR satisfies ϵ -LDP (Theorem 6 in the paper)

Proof Sketch:

- Phase I uses SPL ($\frac{\epsilon}{d}$ per attribute). By **sequential composition**, this satisfies ϵ -LDP [1]
- Phase II perturbs only the pivot using the full privacy budget ϵ . The remaining attributes are generated from the privatized pivot.
 - **Post-processing** [1]: operations on already privatized values incur no additional privacy cost.
- Phase I and Phase II use disjoint user groups, so **parallel composition** preserves overall ϵ -LDP. [2]

[1] Y Zhang, Q Ye, R Chen, H Hu, Q Han, et al. 2023. Trajectory data collection with local differential privacy. Proceedings of the VLDB Endowment.

[2] Frank D McSherry. 2009. Privacy integrated queries: an extensible platform for privacy-preserving data analysis. In Proceedings of the 2009 ACM SIGMOD International Conference on Management of data. 19–30.

Experimental Setup

We evaluate Corr-RR against three baselines using synthetic and real-world datasets.

- Baselines: SPL, RS+FD, RS+RFD (two-phase variant using privatized priors)
- Metric: Mean Squared Error (MSE), lower is better.
- Synthetic Datasets: SynA (star-shaped dependency) and SynB (mixed parent–child dependency)
 - Vary correlation: $\rho \in [0.1, 0.9]$
- Real-world Datasets: Clave, Mushroom, Adult

Results: Impact of Privacy Budget

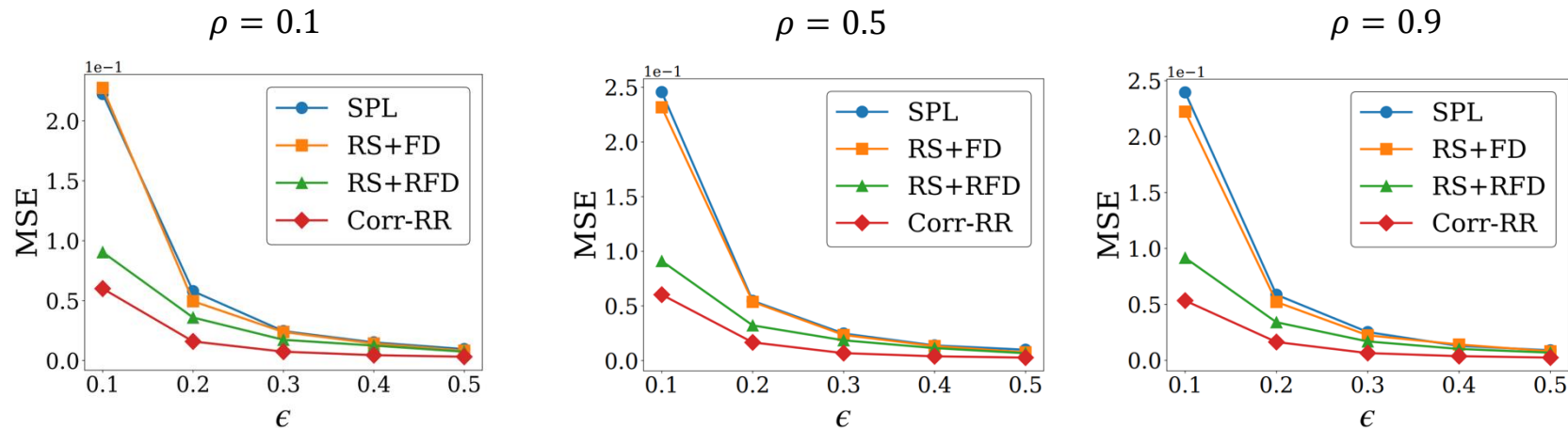


Figure: MSE vs. privacy budget ϵ on SynA with $d = 4$ and $|D| = 4$.

- Corr-RR achieves the lowest MSE across all evaluated ϵ values, providing better utility at the same privacy level.
 - As ϵ increases, the methods converge, but Corr-RR maintains a consistent advantage.

Results: Impact of Number of Attributes

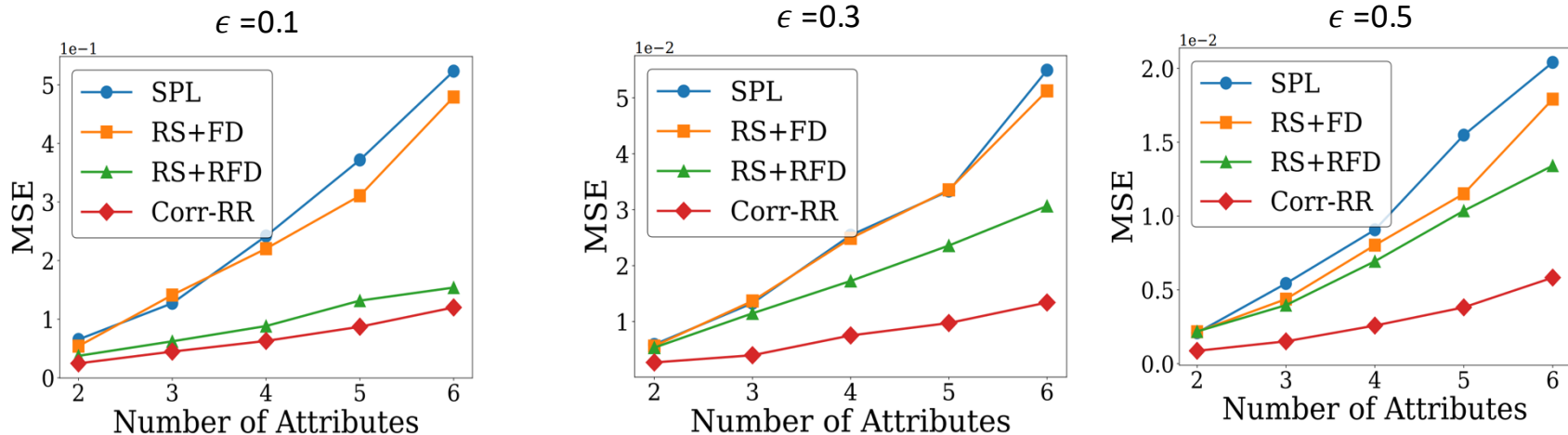


Figure: MSE vs. number of attributes on SynA with $\rho = 0.9$ and $|D| = 4$.

- Baselines degrade sharply as d increases, while Corr-RR scales more effectively with the number of attributes.
 - At $\epsilon = 0.1$, Corr-RR achieves over 4 $\times$ lower error than SPL and a clear advantage over RS+RFD, with the gap widening as d grows.

Results: Impact of Correlation

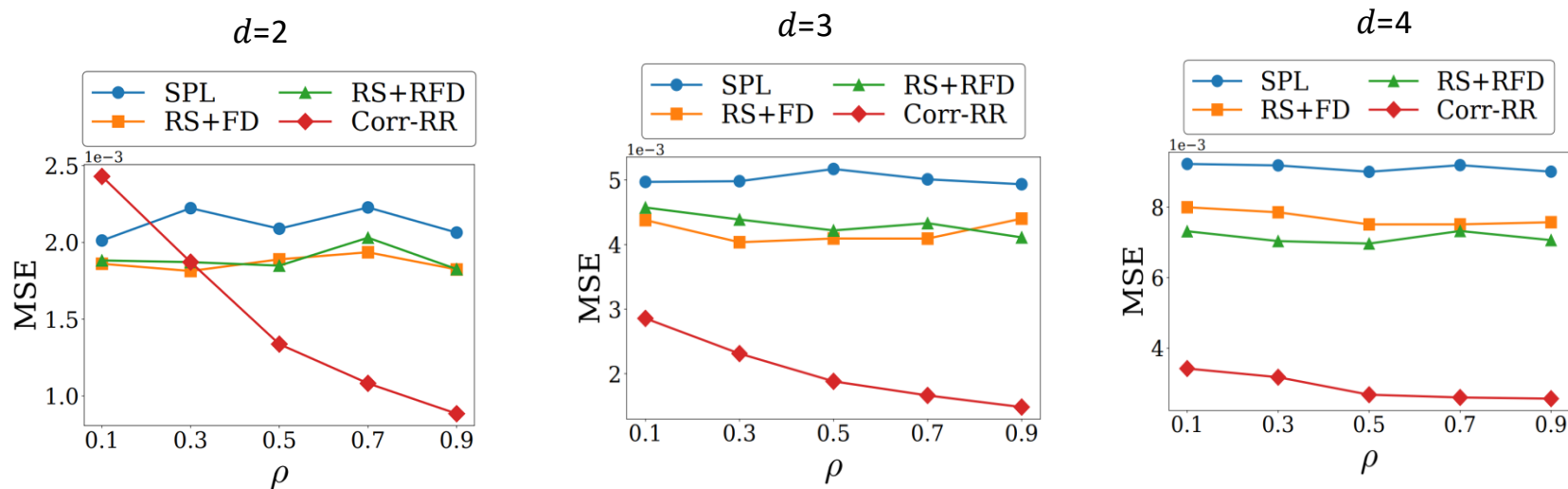


Figure: MSE vs. correlation ρ on SynA. Subplots correspond to different numbers of attributes d .

- Baseline methods remain largely unchanged as correlation increases because they treat attributes independently.
- Corr-RR's error decreases steadily as ρ increases, validating the core idea of **exploiting attribute correlations to improve estimation accuracy**.

Results: Impact of Phase I Subset Size

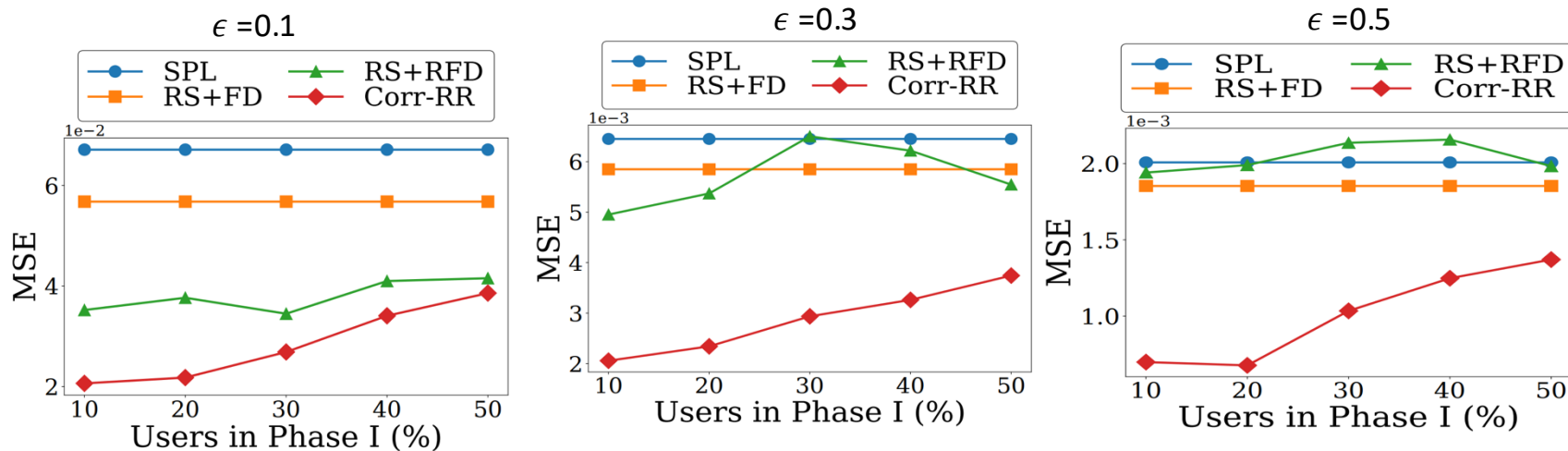


Figure: MSE vs. fraction of users in Phase I on SynA ($n = 20,000$, $d = 2$, $|D| = 4$).

- SPL and RS+FD are single-phase, so they remain unchanged as the Phase I fraction varies.
- RS+RFD (variant) and Corr-RR are two-phase methods, so their error varies with Phase I size.
 - **Corr-RR performs best with a small Phase I fraction (5–10% for large n)** (see Tables 6–9 in the paper), balancing reliable marginals and full-budget reporting in Phase II.

Results: Real-world Data

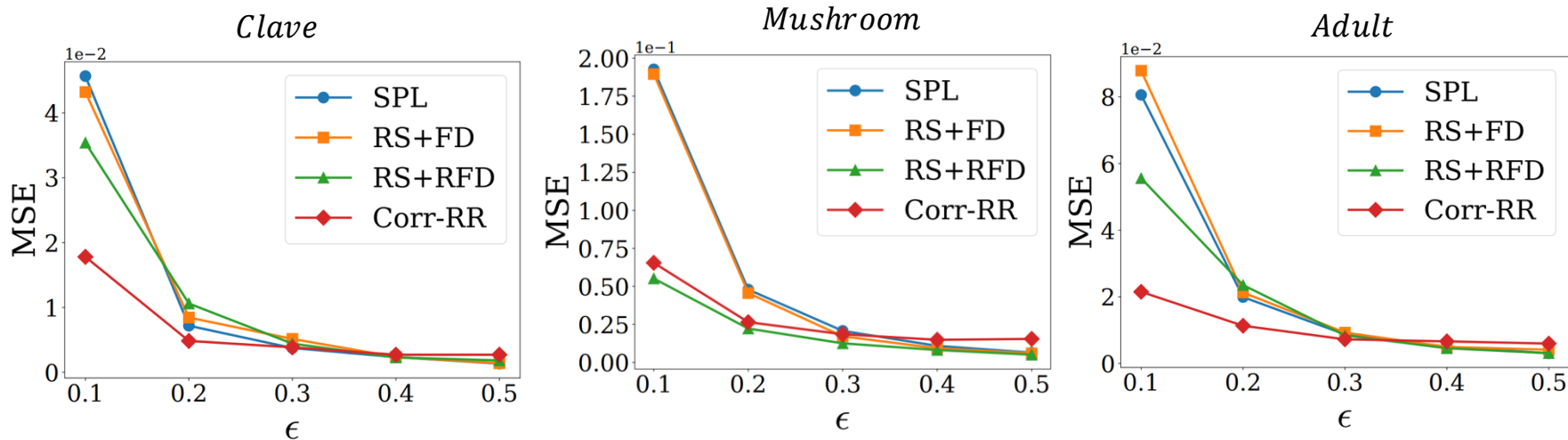


Figure: MSE vs. privacy budget ϵ

- Corr-RR achieves the lowest error on *Clave* and *Adult*, with the biggest gains at small ϵ
- On *Mushroom*, it trails RS+RFD because the dataset has weak attribute correlations and highly skewed marginals. With limited dependency to exploit and a few dominant categories, correlation-guided reuse provides less benefit.

Conclusion

- Existing multi-attribute LDP mechanisms lose utility because they treat attributes independently.
- Two-Phase Privacy Framework:** privately learn dependency signals in Phase I, then use them for correlation-aware reporting in Phase II.
- Under this framework, we introduce **Corr-RR**: perturb one pivot attribute using the full privacy budget ϵ , then generate the remaining attributes by reusing the pivot output with probability p_y ; Corr-RR satisfies ϵ -LDP.
- Corr-RR consistently achieves lower error than SPL, RS+FD, and RS+RFD, with the largest gains in high-dimensional, strongly correlated settings.

Frequency Estimation of **Correlated Multi-attribute Data** under Local Differential Privacy

 Thank You 

RIT | Rochester Institute
of Technology